

SILAS M. FALDE

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EDUCATION

University of Michigan School of Engineering

Junior B.E.S. in Data Science, GPA: 3.13/4.00

Ann Arbor, MI

August 2023 – December 2026

Relevant Courses: Machine Learning, Artificial Intelligence, Computer Vision, Computational Modelling of Complex Systems, Database Management, Regression Analysis, Probability and Statistics, Bayesian Data Analysis, Data Structures and Algorithms

- Michigan Data Science Team (MDST) – Member 2 years
- Heartbeat (A Cappella) – Member 1 year

Western High School

GPA: 3.963

Parma, MI

August 2019 – May 2023

- Men's Varsity Soccer – Captain 1 year, Member 2 years
- Select Choir – Member 3 years

WORK EXPERIENCE

Ally Financial, Inc.

Intern, Data Management

Detroit, MI

May 2025 – August 2025

- Created data dictionaries and data lineage documentation leveraging Python, enhancing data governance and compliance.
- Added data quality checks to existing data pipelines using Python and SQL, improving data integrity and reliability.
- Developed a data availability dashboard in Power BI to monitor data refresh rates and availability, ensuring timely access to critical data.

Michigan Soccer Referee Association

Certified Grassroots Soccer Referee

Ann Arbor, MI

March 2022 – Present

- Officiating on a referee team to create a safe, fair, and fun environment for players, coaches, and spectators.
- Applying knowledge of the rules of soccer to facilitate the sport.

RESEARCH

University of Michigan Department of Epidemiology

Research Assistant – Computational Lead

Ann Arbor, MI

September 2025 – Present

- **Leading** the technical development of a **Python** pipeline using a large language model to classify narratives of collective violence.
- Performing data cleaning, exploratory data analysis, and descriptive statistical analysis on a large dataset using **R** and **Python**.
- Evaluating model performance using **performance metrics** including accuracy, precision, recall, and F1 score.

University of Michigan Department of Epidemiology

Research Assistant

Ann Arbor, MI

September 2024 – May 2025

- Awarded 3rd place (top ~1%) in the CDC Youth Mental Health: Novel Variables Data Challenge.
- Developed **Python** pipelines using a large language model on a high performance computing cluster to extract trends from the narratives of victims of suicide.
- Cleaned, preprocessed, and **feature engineered** a confidential dataset of over 400,000 narratives.
- Applied **natural language processing** techniques to identify relevant cases and enhance the annotation process.
- Documented findings with reports and visualizations to communicate findings to both the research team and the academic community through publication.

PROJECTS

Predicting Survival in the ICU

Project Developer

Ann Arbor, MI

January 2025 – March 2025

- Developed logistic regression and RNN models to predict survival of patients in the ICU using a public dataset using **Scikit-learn** and **PyTorch**.
- Performed data cleaning, preprocessing, and feature engineering to prepare the data for model training.
- Fine-tuned and evaluated the models using **cross-validation** and **grid search**.

SKILLS

Programming

C++, Python, R, SQL, Java

Python

PyTorch, Scikit-learn, NumPy, Pandas, Matplotlib, NLTK, spaCy

Software

Git, VS Code, Microsoft Office Suite, Google Suite, Snowflake, Collibra, Alation, Manta